the maturity of the bond). And both of them are to a significant extent driven by the same variable, i.e. the credit quality of the bond issuer (see Chapter 12). It is thus natural to compare both numbers – which introduces a relationship between asset and basis swaps on one side and CDS on the other side. As a result, we obtain an RV relationship between both and are able to extend the global links and mutual influences between all asset and basis swaps also to and with CDS.

Concerning the RV relationship between the USD asset swap spreads of basis swapped bonds ("USD ASW") and the CDS of the bond issuer (with the length of the CDS equal to the maturity of the bond), there are two possible approaches:

- Aiming for an arbitrage *equality*, one can combine the pricing models discussed in the last chapters and extract the pure credit risk of the bond issuer from both the USD ASW and the CDS. While a strong theoretical case can be made for the equality of the same variable contained in two different market quotes, applying this concept in practice faces the issue of too few observations for a precise estimation of the DO (delivery option) and FX component also contained in the CDS quotes.
- Aiming for a relationship which can be more easily implemented in practice, the arbitrage *inequality* USD ASW (market observed, not adjusted) $\leq$ CDS (market observed, not adjusted) can be used. Since this relationship is based on qualitative arguments only, such as the positivity of the DO and FX component, it is independent of their estimation error. This approach could be characterized as a tactical retreat from an arbitrage equality to an inequality in order to express the concept in terms which can be applied to the market independent of modeling issues.

The first two sections of this chapter discuss the two relationships imposed on bond and swap markets,

1. Between USD ASW and local ASW via the CCBS (and ICBS).
2. Between USD ASW and the CDS.

The first relationship is simply given by the formula for calculating USD ASW from local ASW and the basis swaps, the second relationship compares USD ASW and CDS as two different market prices, which contain a common element, i.e. the credit quality of the bond issuer.

After treating both relationships separately, they will be combined into a single formula linking four markets, local ASW, USD ASW, CCBS (and ICBS), and CDS. This formula will then be applied to a number of case studies, explaining the different equilibria observed for bonds of higher and lower credit quality.

Given the number of variables connected, there is usually more than one possible outcome of changes in a particular variable. For example, a richening of Bunds versus EURIBOR can result in the USD–EUR CCBS absorbing this richening, in which case the balance between all CCBS is disturbed and will readjust, and/or in Bunds also richening versus USD SOFR, in which case the balance between all (basis swapped) bonds is disturbed and will readjust. The complexity of these mutual influences (similar to the three-body problem in astronomy) corresponds to the complexity of the relationships. Fortunately, the arbitrage inequality provides a clear and hard boundary.

CALCULATING USD SWAP SPREADS FOR FOREIGN BONDS

Recall that the cross-currency basis swap (CCBS) allows a USD-based institution to invest in any product independent of its currency without FX risk. For example, it could enter a JPY CCBS, use the JPY principal it initially receives, buy a JGB, asset swap it, and thereby create JPY TONAR + ASW cash flows. With these cash flows, it can serve the running CCBS payments, receiving USD SOFR (plus a spread). If the maturity of the CCBS equals the maturity of the JGB, it will receive the JPY principal repayment from the bond at the same time at which the final principal exchange in the CCBS will convert it back into the original USD principal. Note that, though it has created the cash flows by means of a JGB investment, through the CCBS it deals in the end only with USD cash flows. Of course, any other currency could be used as a basis for comparison as well, and a EUR-based investor is able to express and evaluate all bonds globally as a spread versus EURIBOR.

The importance of this statement (apart from the convenience it offers to USD-based investors) is that, through combining asset and basis swaps, all bonds can be reduced into a single number and can therefore be compared relative to the same benchmark. The mathematics of this expression is straightforward under some simplifying assumptions such as par bonds, which will be relaxed in Chapter 17. The cash flow in foreign currency is the foreign reference rate (e.g. TONAR) + foreign asset swap spread. Thus, after payment of foreign reference rate + CCBS in the CCBS, one receives USD SOFR (from the CCBS) and has the foreign swap spread minus the CCBS premium left in foreign currency. This then needs to be converted into USD by adjusting for the difference in basis point value (BPV).

Thus, the USD asset swap spread ("USD ASW")[1] is:

$$\frac{Swapspread_{foreign} - CCBS}{CF},$$

[1]Sticking to the convention "Reference Rate + X" for swap spreads.

where CF denotes the conversion factor, that is

$$\frac{BPV_{USD}}{BPV_{foreign}}$$

(with the BPV of the swap rates of the relevant maturity and with the relevant reference rate).

In case the domestic asset swap uses a different reference rate than the CCBS, the appropriate ICBS needs to be added. For example, if the JGB is asset swapped in TIBOR, it first needs to be converted via an ICBS into TONAR, which is then exchanged via the CCBS into SOFR. Correspondingly, before applying the USD ASW formula above, the JPY TONAR ASW needs to be calculated by adding the ICBS to the TIBOR ASW. This TONAR ASW – in general terms, the swap spread against the non-USD reference rate used in the CCBS – gives then the input variable $Swapspread_{foreign}$ for the USD ASW formula. For simplicity, the formula and the rest of this chapter assume that the reference rate of the CCBS is the same as the reference rate of the domestic swap; if this is not the case, the variable "CCBS" needs to be adjusted with the relevant ICBS like in the TIBOR example.

As explained at the end of Chapter 15, despite the transition to SOFR, we'll continue to use historical case studies with USD LIBOR as reference rate in order to avoid structural breaks in the CCBS time series. Before the transition from LIBOR to SOFR as main US reference rate, (3M) USD LIBOR played the role as global benchmark and hence as USD leg in CCBS. The general formula from above covers all reference rates and can also be used to calculate the USD ASW versus LIBOR. For example, the ASW of a JGB versus 6M JPY LIBOR is translated via the 3M/6M JPY LIBOR ICBS into a 3M ASW and then taken as input into the formula. Using the data from Sep 23, 2012, the 10-year (10Y) JGB benchmark was priced to have a 6M JPY ASW of -1 bp. Adding the 3M/6M JPY ICBS of 12 bp gives a 3M JPY ASW of 11 bp. With the CCBS at -64.5 bp and a CF of 0.95,[2] the equation above returns a spread of 79 bp above USD LIBOR for the 10Y JGB.

A useful graphical representation of the way global bonds compare relative to the universal yardstick of USD SOFR is to display the USD swap spreads of the relevant bonds as a function of their maturities. Figure 16.1 shows the USD (3M LIBOR) swap spreads for US Treasuries, basis swapped Bunds, and basis swapped JGBs. Such a chart can serve as starting point for global investment (and funding) decisions.

Note that while Bunds track the US Treasury curve (on a USD LIBOR basis) at a rather constant and low spread, JGBs are priced at a significantly

[2]Calculated by dividing the BPV of a 10Y USD swap by the BPV of a 10Y JPY swap.

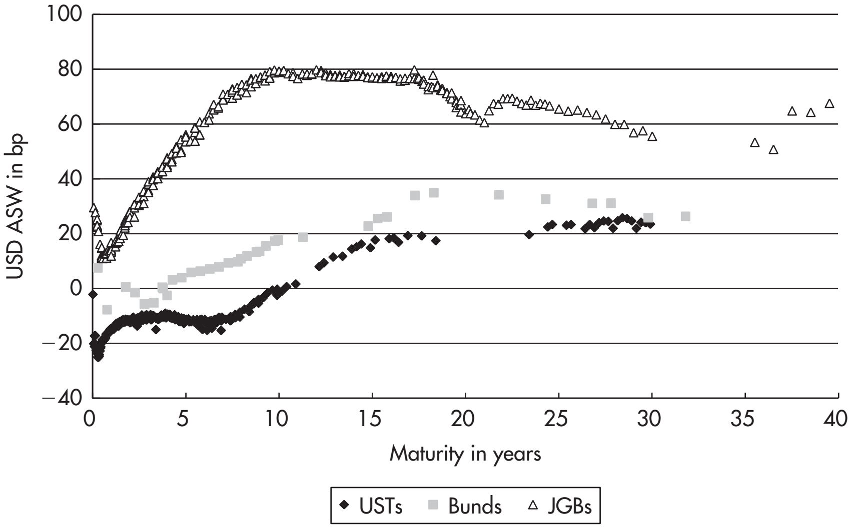

FIGURE 16.1 USD LIBOR swap spreads of US Treasuries, Bunds, and JGBs as of 23 Sep 2012.
Sources: data – Bloomberg; chart – Authors.

Data period: "Current" market data as of 23 Sep 2012.

higher USD ASW and follow an independent path. A comprehensive explanation for this behavior will be attempted at the end of this chapter. But already the equation above offers a mathematical explanation for the high USD ASW of JGBs based on the relative stability of the JPY ASW of JGBs (which will be explained later on): as the banking crisis resulted both in a higher 3M/6M JPY LIBOR ICBS and in a more negative USD–JPY CCBS, the USD ASW of JGBs needed to increase.

THE EQUILIBRIUM BETWEEN ASSET AND BASIS SWAPS

As the CCBS exists, there exists a USD ASW in addition to the local ASW and a relationship between the two. In particular, this relationship is given by the formula[3]:

$$USDASW = \frac{EURASW}{CF} - \frac{CCBS}{CF}$$

[3]For simplicity, this formula and discussion assume that the reference rate of the local asset swap is the same as the reference rate of the CCBS. If this is not the case, the variable CCBS needs to be adjusted with the relevant ICBS, as explained above.

Hence, any change in one of the constituents affects all others, though there are different ways in which the equilibrium can be maintained. For example, a richening of Bunds on a EUR ASW basis could be compensated either by a widening (more negative) of the EUR CCBS or by a richening of Bunds on a USD ASW basis.

As the equation does not determine the way a change in the USD ASW would be reflected by changes in the local ASW and CCBS, we need to leave the realm of pure mathematics and consider actual market data.

As a starting point for a brief empirical study of the way this equilibrium works in practice, we depict the evolution of the USD ASW of 5Y US Treasuries, 5Y Bunds, and 5Y JGBs in Figure 16.2. The USD ASW of 5Y Bunds followed the evolution of the USD ASW of 5Y US Treasuries at a rather constant spread. Actually, the correlation between the two time series is 0.73. This means that, for Bunds, the existence of a global yardstick (the USD ASW) links the pricing of Bunds to the pricing of US Treasuries. Therefore, the USD ASW level of US Treasuries as well as the CCBS should have a significant impact on the pricing of Bunds denominated in euros.

On the other hand, the USD ASW of JGBs has appeared more independent. This could reflect the fact that JGBs are considered riskier than Bunds and hence are not as seen as close alternatives for US Treasuries. As the influence of the global asset selection process illustrated in Figure 16.1 is greater if

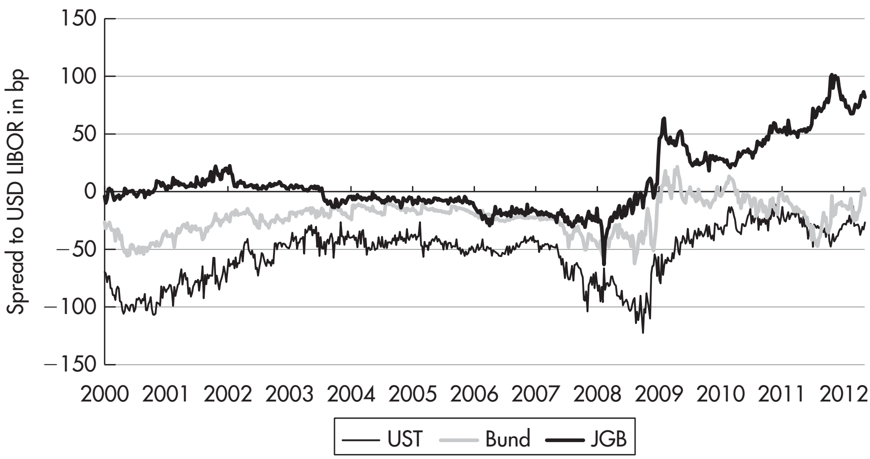

FIGURE 16.2 USD swap spread history of the 5Y US Treasury, Bund, and JGB.
Sources: data – Bloomberg; chart – Authors.

Data period: 8 Feb 2000 to 12 Jun 2012, weekly data.

the bonds have a comparable credit standing, the USD ASW of US Treasuries has a significant impact on the pricing of low-risk bonds. Risky bonds, by contrast, have more scope to follow their own USD ASW pricing. This leads to the issue of the determinants of the USD ASW of JGBs. And as the USD ASW of JGBs is not tied strongly to the USD ASW of US Treasuries, it is conceivable that the existence of the USD ASW exerts less influence on the pricing of JGBs in yen.

From Figure 16.2 it is obvious that the same equation manifests itself differently in different markets and that for high-risk bonds an important driving force is still missing. This driving force is the credit quality as measured by the (adjusted) CDS. We will now add this element to the links between the swap markets and after discussing the arbitrage relationships revisit the case studies in order to observe the impact of the CDS on the equilibrium.

ARBITRAGE EQUALITY BETWEEN USD ASW AND CDS

The combination of asset and basis swaps[4] is conceptually a reduction of every bond to a spread over USD SOFR. Credit default swaps (CDS) follow precisely the same concept. Every bond globally is evaluated by a CDS in the form of a spread (the CDS premium) over USD SOFR. Moreover, the source of that spread is in both cases to a large extent driven by the credit quality of the bond. The perception of poor credit quality results in a large USD ASW and in a large CDS.

The swap spread model presented in Chapter 12 reflects the link between ASW and CDS by incorporating the adjusted CDS level as key component. In fact, SOFR–ASW of US Treasuries have been modeled by the adjusted CDS plus the expected specialness: treasuries without specialness should therefore quote at a SOFR–ASW equal to the adjusted CDS (abstracting from other driving forces like haircuts). While the funding swap becomes more influential for other types of swap spreads, e.g. LIBOR–ASW, the CDS remains an important component in all of them, which explains both the term structure and the cyclicality of swap spreads to a large extent.

Let us now assume that both the USD ASW of a bond and its CDS are driven by the credit quality of the bond not only *to a large extent* but *exclusively*. Of course, this is an artificial assumption, which allows us to outline the general concept of no-arbitrage models for CDS pricing in clean and easy terms. After the assumption has fulfilled this function, we shall describe the way the

[4]In this section, the term "basis swap" (BSW) refers to the cross-currency basis swap (CCBS), adjusted for the intra currency basis swap (ICBS) where necessary.

general concept needs to be modified in order to work in the real world, where both the USD ASW and the CDS provide the information about credit quality in different forms, overlying that information with other elements.

Under that assumption, *the combination of asset&basis swaps can be considered as a synthetic short position in a CDS on the bond issuer*.

- The risk in the combination asset&basis swap[5] as well as in a short CDS is the default of the bond issuer.
- For taking that risk, an investor is compensated with a spread above USD SOFR, in case of asset&basis swaps with the USD swap spread of the bond, and in case of a short CDS with the default swap premium.

Since both the USD ASW of a bond and the CDS of the issuer of that bond reflect the credit risk in terms of a spread versus USD SOFR, it is natural to compare the two. This comparison allows detection of relative value opportunities between the two markets, i.e. the investigation whether the credit risk is reflected in the bond and CDS markets *relative to each other* in a consistent way. If it is not, trades between the USD ASW and the CDS can exploit the different assessment of the same credit risk in both markets, without being exposed to the level of credit risk or to a default.

If the USD swap spread of a JGB was higher than its CDS, we could asset&basis swap the JGB and buy default protection on Japan, realizing the difference between the two spreads versus USD SOFR, the USD ASW of the JGB minus the CDS, as risk-free arbitrage[6] profit: until a default occurs (and until the maturity of the bond if no default occurs), we obtain the difference as profit. And in a default, the CDS payment covers our losses from the bond.[7]

Put otherwise, by combining the three swaps (asset, basis, and default), we have step-by-step reduced the risks involved in a bond investment, as illustrated in Table 16.1.

As the combination of all swaps results in a risk-free position, there should be no compensation for taking that position. Thus, as ASW&BSW&CDS = 0, the cash flow from asset&basis swapping a bond (its USD swap spread) should compensate for its default protection (i.e. the USD ASW

[5]By the symbol "ASW&BSW," we refer both to the action of asset and basis swapping a bond and to its result (i.e. the USD swap spread of that bond). Note that while the actions and the symbol "&" are additive, the USD swap spread is given by the difference (ASW-BSW)/CF.

[6]This arbitrage requires the artificial assumption made above. Later, we will discuss how to apply that arbitrage concept also in the real market, where the assumption does not hold, which may require us to drop the term "risk-free arbitrage" in its strict sense.

[7]Abstracting from the issue of the ASW and BSW remaining after default.

TABLE 16.1 Risk Exposure of a Bond Together with Different Combinations of Swaps

Position	**Exposed to**		
	Yield risk[1]	**FX risk**	**Default risk**
Bond	Yes	Yes	Yes
Bond&ASW	No	Yes	Yes
Bond&ASW&BSW	No	No	Yes
Bond&ASW&BSW&CDS	No	No	No

[1]Note: this table should be understood in broad conceptual terms. In particular, as the duration of an asset-swapped bond position is very low, we enter "No" in the yield risk column, even though, strictly speaking, it has some P&L, depending on the level of short rates.

should equal the CDS). This is an alternative way to reach the conclusion from above, that there is an arbitrage relationship between asset&basis swapped bonds and CDS. Also note that while analyzing the ASW, the BSW, and the CDS in isolation, there are many driving factors and complex relationships between them to consider, when putting them all together, one arrives at a simple arbitrage relationship that should hold independently of the specific issues and flows that drive the individual swaps

Thus, we have intuitively derived the equation USD ASW = CDS of the well-known arbitrage models for CDS pricing and trading (with the difference between CDS and USD ASW being called "basis"). In reality, however, this framework is just the starting point for further and deeper analysis. Since the derivation of the equation USD ASW = CDS required us to artificially exclude all other elements except credit information from both USD ASW and CDS, we need now to consider how to deal with these additional parts of both sides.

If the goal is to maintain the arbitrage *equality*, one can use the models presented in the previous chapters in order to isolate the pure credit information from both the USD ASW and the CDS quotes observed in the market.

Specifically, starting from the USD ASW, one can use the model outlined in Chapter 12 and its subsequent extensions to abstract from non-credit related elements. The funding component can be isolated by directly applying the formula from Chapter 12:

Adjusted CDS = USD ASW (observed in market) – Fair value of the basis swap between repo and the reference rate of the swap.

In case of SOFR–ASW, the funding swap only covers the specialness and is usually rather small, in which case the approximation *Adjusted CDS ≈ USD SOFR–ASW* might be acceptable. From that starting point, the additional driving forces of swap spreads not included in the model and discussed in

Chapter 18, such as haircuts, can also be subtracted from the observed USD ASW, just like the funding component. The idea is to "adjust" the USD ASW given by the market by excluding all elements not related to the credit quality of the bond, until only the component *Adjusted CDS* remains.

And also starting from the CDS quotes, using the evaluation of the DO and FX component from Chapter 13, all elements not related to the credit quality of the bond can be excluded, until only the component *Adjusted CDS* remains:

Adjusted CDS = CDS (observed in market) – value of DO – value of FX component.

Once these exercises are completed, one can then express the arbitrage equality as:

Adjusted CDS extracted from USD ASW = Adjusted CDS extracted from CDS quotes,

i.e. argue for the equality of the prices for the 'pure' credit risk by making sure that both sides of the equation only reflect 'pure' credit risk.

Hence, the two hurdles on the way to an arbitrage equality are the two adjustments: once these are made and two measures for the pure credit risk of the same bonds are obtained from the USD ASW quotes, on one side, and from the CDS quoted, on the other, a strong case can be made for their equality.

- Regarding the adjustment of USD ASW, before the transition it involved removing the secured–unsecured basis from USD ASW versus LIBOR and hence dealing with the variable "cost of capital" and the problem of it being unobservable ex ante (Chapter 11) as well as its correlation to the sovereign credit risk. Fortunately, this problem (that occupied many pages of the first edition) does not exist for USD ASW versus SOFR anymore, which do not involve the secured–unsecured basis. The remaining non-credit-related elements in the USD ASW are mainly the specialness and haircut financing costs, which are easier to estimate (Chapters 12 and 18). Seen from this perspective, the transition from LIBOR to SOFR has been a big step toward the arbitrage equality.
- Regarding the adjustment of CDS, on the other hand, the problems mentioned in Chapter 13 remain: Given the low number of observations, the estimation of the DO and FX component in a CDS is subject to a significant error.

In the current situation, extracting the pure credit information from SOFR swap spreads seems already feasible (in case of a sufficient database for repo specialness), while the estimations adjusting CDS quotes are still in a wide range. In case this range can be sufficiently narrowed due to more future observations, also the second hurdle toward the arbitrage equality can be taken.

ARBITRAGE INEQUALITY BETWEEN USD ASW AND CDS

While exact quantifications of the DO and FX component in the CDS remain difficult, the qualitative statement that both of them should be positive is much easier:

- The statement DO ≥ 0 is a mathematical truth as no option can have a negative value.
- By contrast, the statement FX component ≥ 0 requires the economic assumption that the currency of a defaulting country weakens. Unlike the DO, the FX component is not an option, but a short position in the FX rate of a defaulting country conditional on a default and therefore exposed to both a weakening and a strengthening.

With these two statements, *adjusted CDS* $\leq$ *CDS* holds true, and thus it is easy to derive from the arbitrage equality

> *Adjusted CDS extracted from USD ASW = Adjusted CDS extracted from CDS quotes*

the arbitrage inequality

> *Adjusted CDS extracted from USD ASW* $\leq$ *CDS quotes (unadjusted, market observed).*

Turning to the left side of the equation, we have modeled the SOFR–ASW by adding the expected specialness (a non-positive number) and the haircut costs (a non-negative number) to the pure credit element captured by the adjusted CDS. In case of no specialness and haircuts, the approximation *Adjusted CDS* $\approx$ *USD SOFR–ASW* might be acceptable for SOFR–ASW.[8] in which case the arbitrage inequality simplifies to

> *USD ASW (unadjusted, market observed)* $\leq$ *CDS (unadjusted, market observed),*

i.e. two market observed variables without any risk of misestimation or model errors.

By sacrificing one side of the equation (and making some assumptions), one can thereby get rid of any dependency on models, observation and estimation problems and obtain a relationship between traded variables. The ability to execute the arbitrage inequality directly in the market contributes to

[8] Unlike for LIBOR–ASW, which still contain the secured–unsecured basis. This should be kept in mind for the historical case studies below.